The empirical recurrence for OEIS A302639, recovered from its tail

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Abstract

OEIS A302639 records "Empirical recurrence of order 81" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 253$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 87$. Its last coefficient is $c_{81} = 96 \neq 0$, so no further tail satisfies anything shorter; any order-81 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302639 is "Number of 6Xn 0..1 arrays with every element equal to 0, 1, 3 or 6 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$32, 42, 82, 248, 719, 2262, 6460, 19799, \ \ldots$$

The entry states:

> Empirical recurrence of order 81 (see link above)

As of the "Last modified" line on the live entry (Apr 10 2018 19:35:28, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 253$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 253$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 81: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 81.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 506$ exact terms from $n = 6$ on, it returns order 81 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{81} c_i\, a(n-i),$$

c_1	1	c_{22}	-225150	c_{43}	-1557216	c_{64}	-105282
c_2	2	c_{23}	20736	c_{44}	19941837	c_{65}	148728
c_3	5	c_{24}	-2157646	c_{45}	-8987951	c_{66}	94806
c_4	57	c_{25}	686221	c_{46}	-715322	c_{67}	54278
c_5	-38	c_{26}	671358	c_{47}	-414268	c_{68}	13964
c_6	-73	c_{27}	-355162	c_{48}	-13707720	c_{69}	-32792
c_7	-155	c_{28}	5841770	c_{49}	7023230	c_{70}	-29892
c_8	-1261	c_{29}	-1867346	c_{50}	29942	c_{71}	-4732
c_9	639	c_{30}	-1467674	c_{51}	1631292	c_{72}	-3424
c_{10}	1098	c_{31}	1231367	c_{52}	6997085	c_{73}	7032
c_{11}	1783	c_{32}	-11903829	c_{53}	-4139151	c_{74}	5240
c_{12}	15153	c_{33}	3983279	c_{54}	8284	c_{75}	-168
c_{13}	-6306	c_{34}	2322371	c_{55}	-1535513	c_{76}	880
c_{14}	-9548	c_{35}	-2371558	c_{56}	-2567645	c_{77}	-1184
c_{15}	-9983	c_{36}	18477727	c_{57}	1816086	c_{78}	-400
c_{16}	-114398	c_{37}	-6688809	c_{58}	148449	c_{79}	48
c_{17}	41409	c_{38}	-2573135	c_{59}	827101	c_{80}	-96
c_{18}	55075	c_{39}	2733772	c_{60}	644140	c_{81}	96
c_{19}	24797	c_{40}	-21959026	c_{61}	-591662		
c_{20}	587715	c_{41}	8803179	c_{62}	-172242		
c_{21}	-194661	c_{42}	1856811	c_{63}	-276424		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 87$, and it is the recurrence OEIS A302639 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{81} - c_1 x^{81-1} - \cdots - c_{81}$. Its constant term is $-c_{81} = -96$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 81 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 81, so $\deg q = 81$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 81, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 81, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 96.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-81 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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